IoT-Enabled Smart bus route monitoring and revenue management system

Project ID

R26-ISE-004

Project Proposal Report

Sachintha H D D – IT 22264152

Component: Policy- Driven Bus Scheduling and Time-Slot Management Engine

B.Sc. (Hons) Degree in Information Technology Specialized in Information System Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Sachintha H D D	**IT 22264152**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Senjeewa Perera	19/2/2026	

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ABSTRACT

Public bus transport systems often experience operational difficulties because they need better scheduling systems, and they track route compliance poorly, and their service period management needs improvement. Bus operations on many routes still depend on timekeepers who manually control their bus schedules, which creates challenges for adjusting schedules during delays, breakdowns, and service interruptions. The system enables transport authorities to face difficulties when operational policies need to be enforced because it leads to uneven bus distribution during different time periods and peak-hour demand management problems. The existing system requires the development of an intelligent, automated transport monitoring solution that addresses current system limitations.

The project solves the problem through its IoT-based smart bus route monitoring system, which tracks buses using GPS technology and verifies their route compliance while managing operations through automated processes. The system uses the Policy & Time-Slot Engine to control transport policies and classify bus operations into predefined service periods, which include peak hours and normal service periods, and night services. The engine assesses operational data to determine compliance with established policy rules and enables the system to modify schedules automatically during disruptive events. The system ensures bus distribution remains equitable while enabling policy adherence and boosting operational efficiency.

The system will use IoT-based GPS devices, cloud-based processing, rule-based policy engines, and web and mobile applications as its core technologies to perform monitoring and decision-making tasks. The expected outcomes of this project will result in better service reliability and automatic policy enforcement, and reduced need for staff-run timetable operations. The system brings social benefits because it enhances public transportation service reliability and availability. The system enables transport authorities and bus operators to monitor their operations while managing their policies through a scalable solution that enhances public transport system efficiency.

Keywords: Public Transport Monitoring, Policy-Based Scheduling, Time-Slot Engine, IoT Transport Systems, Route Compliance Monitoring.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor and co-supervisor for their valuable guidance, continuous support, and constructive feedback throughout the development of this project. The academic staff of the Faculty of Computing at the Sri Lanka Institute of Information Technology (SLIIT) provided academic support to me during my research work while sharing their knowledge with me, which I appreciate. My appreciation also goes to the Computer System Administrator of the Road Passenger Transport Authority for providing useful information and insights related to the existing transport management practices. My friends and family supported me through this work which I want to express my gratitude.

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LIST OF ABBREVIATIONS

Abbreviation	Description
API	Application Programming Interface
DB	Database
ETA	Estimated Time of Arrival
GPS	Global Positioning System
IMU	Inertial Measurement Unit
IoT	Internet of Things
ITS	Intelligent Transportation System
REST	Representative State Transfer
UI	User Interface
RPTA	Road Passenger Transport Authority

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1. INTRODUCTION

1.1 Background and Context

Public bus transportation in urban and suburban areas serves as a vital service that enables thousands of people to travel each day. In Sri Lanka's Western Province, private bus operators control most of the public transport system because they must operate according to approved routes and schedules, which transport authorities establish. The organization needs help to achieve its main objectives because it faces difficulties with both delivery service and route tracking operations.

The timekeepers continue to use manual methods for running bus operations because they use their available time to create weekly schedules and track bus movements. The manual process cannot handle unexpected events, which include delays, bus breakdowns, and irregular service patterns that disrupt operations. The system causes buses to operate with inconsistent service throughout the day, which results in improper peak-hour demand handling and extended passenger waiting periods with unreliable service delivery.

Transport authorities and bus operators experience operational difficulties because they lack systems that automate monitoring and enforce rules. Authorities struggle to monitor whether buses follow approved schedules, while operators have limited mechanisms to demonstrate compliant operations or receive incentives for reliable service. The organization needs to establish a system that will enable operation tracking, together with the creation of scheduling policies that support its operational success.

1.2 Literature Review

Public bus scheduling is a long-standing problem that deals with how to put the vehicles and drivers on the right trips while keeping all the rules in check. In the literature, people usually split public transport scheduling into linked tasks. The Vehicle Scheduling Problem (VSP) is about deciding which vehicle goes on which scheduled trip. The Driver Scheduling Problem (DSP) is about figuring out each driver's duties, making sure they do not work too many hours and that they get enough rest. Driver rostering takes the ideas but stretches them over longer periods, such as a week or a month, and is called the Driver Rostering Problem (DRP) [1].

1.2.1 Vehicle and Driver Scheduling

Vehicle scheduling aims to ensure that all planned trips are covered by an appropriate set of buses while minimizing operational costs and satisfying vehicle-related constraints. Mesquita et al. [1] identify vehicle scheduling and driver scheduling as strongly interdependent components of public transport planning. A change in vehicle allocation can consequently affect driver duties and overall operational feasibility. Similarly, Shen et al. [2] reviewed driver scheduling approaches and highlighted the use of mathematical programming and intelligent optimization techniques for constructing feasible driver schedules under operational and labor constraints.

Recent reviews show that mathematical optimization is still a key method for planning public transport. Studies about transit timetables and schedules often use Mixed-Integer Linear Programming (MILP) to set rules for operations.. People are using other methods, like heuristics, metaheuristics and combinations of these to handle big problems and difficult calculations [3]. Still methods based on optimization can take a lot of time and resources when the scheduling problem needs to be solved times because things change during the day.

This matters a lot for the system we are working on because buses can't be assigned without checking if drivers are available. The scheduling part of the system looks at both buses and drivers when making plans. A resource has to meet rules like being available, having rest and not driving too long before it can be used for a trip.

A recent literature review of transit network timetabling similarly identified real-time schedule adjustment as an important area requiring further development. The review found that although optimization techniques are widely applied, integrating real-time scheduling adjustments with operational requirements remains a research challenge [3].

1.2.3 Real-Time Rescheduling under Bus Breakdowns and Disruptions

Unexpected vehicle breakdowns are a problem for people who rely on public transportation. When a scheduled bus stops working while it is still running, the rest of the fleet has to change routes to pick up the passengers that were supposed to travel on the bus and on the following trips that were planned for that bus.. Colleagues [6] called this the Vehicle Rescheduling Problem (VRSP). In their study, they looked at how much it costs to run the buses, how much it costs when the schedule is messed up, and how much it costs when trips have to be cancelled. Their results show that vehicle breakdowns can force the transit team to change the fleet plan right away. They also suggest a relaxation-based approach that helps solve the rescheduling problem quickly.

Further research has expanded this idea by looking at changing the timing of trips. Li et al. [7] introduced a vehicle rescheduling problem that includes changing the time of trips. In this approach, some trips can be delayed. This helps to make the schedule more flexible. It also helps to reduce the number of trips that have to be canceled after a vehicle breaks down. This shows that handling disruptions can involve more than replacing a broken vehicle. It can also mean changing when services happen.

Another example is from Lai and Leung [8]. They studied real-time rescheduling and handling disruptions. They used a system that combines vehicle scheduling and crew scheduling. Their method uses information that is happening now. It also uses information. This helps to adjust schedules when travel times are not certain and change over time. They created a rolling-horizon method. This method keeps updating the schedule as things on the ground change.

Real-time rescheduling has been studied a lot from an optimization point of view. Yet many methods treat disruption management as a mathematical optimization problem aiming to produce the almost best recovery plan. In practice, connecting real-time event detection, checking resources, applying rules, reassigning trips automatically, and then regenerating the schedule is not highlighted as a complete operational workflow.

1.2.4 Heuristic and Greedy Approaches, for Bus Resource Allocation

Public transport scheduling problems can become very hard for computers. In those cases people often turn to methods and greedy methods. Heuristic methods and greedy methods give solutions in a reasonable amount of time. In reviews of driver scheduling research many different solution methods appear. Solution methods include integer programming, genetic

algorithms, tabu search, ant-colony methods, and other types of optimization techniques [2].

For scheduling systems, a greedy approach can be helpful when having a feasible resource allocation right away is more important than getting the best possible solution overall. Instead of checking every possible combination of buses, a greedy method picks one available resource at a time based on set priority rules. After picking it updates the resource's status and moves on to the trip.

In the system we are proposing, the scheduling method is split based on the time horizon. Daily scheduling uses a rule-based and constraint-driven way to pick resources. Buses and drivers are filtered first by rules and only then is an eligible resource chosen. Monthly schedule generation, on the hand, uses a greedy allocation method. It picks buses that have the trips already assigned and, when possible, those that are available the earliest. This difference matters because the two scheduling tasks do not use the same way of allocating resources.

1.2.5 Disruption-Aware Dynamic Schedule Adaptation

In studies of public transport, researchers are increasingly pointing out that schedules must be able to handle unexpected operational disruptions instead of being thought of as unchangeable plans. Gkiotsalitis and Cats examined real-time public transport control methods. Gkiotsalitis and Cats found that rescheduling, dynamic dispatching, and other similar control measures can help lessen the impact of service variability and disruptions. Gkiotsalitis and Cats also point out the practical difficulties that arise when trying to use these methods in real-time operations.

Recent studies have focused on disruption scenarios and bus timetable resilience. Shen et al. [10] point out that better disruption models are needed and that disruption handling must work closely with timetable optimization, especially when several disruption conditions impact public transport operations.

These studies show that dynamic rescheduling and disruption management are research areas.. There still exists an opportunity to create a practical event-driven scheduling framework where an operational event automatically alters the state of the fleet and triggers new scheduling decisions instead of making the whole process rely on manual schedule reconstruction

1.3 Research Gap

The literature shows that public transport scheduling has been studied a lot using vehicle scheduling, driver scheduling, math optimization, heuristics and real-time disruption handling. Researchers have also looked at vehicle breakdowns. How to reschedule the other vehicles. So the missing piece, in this study is not that bus breakdown rescheduling does not exist. The missing piece is how to bring real-time events into a full automated scheduling workflow that always updates which resources are free and then builds new schedules when the fleet changes or when rules shift.

Existing vehicle scheduling studies mostly concentrate on matching buses to fixed trips while research on driver scheduling focuses on building work schedules for drivers that follow labor rules and operational limits [1] [2]. Studies on scheduling go further by including real-time changes in operations [4] [5]. There is also work, on rescheduling vehicles showing how the remaining fleet can be reassigned after a breakdown occurs [6] [7].. Despite all these efforts each area tends to be seen as its own separate problem. They are not usually connected into one process that starts with detecting an event then updates the state of the fleet filters available resources, assigns trips again and finally rebuilds the schedule going forward.

The proposed research tackles this gap by introducing an event-driven scheduling system that works with real-time bus monitoring. When a bus breaks down or an emergency happens the affected bus is quickly marked as unavailable. The scheduling engine then takes note of the fleet status and checks the remaining buses and drivers against set operational rules. It finds resources and tries to reassign trips that were affected. If there are not enough buses or drivers to keep all trips running as planned, the system identifies which trips can't be supported and adjusts the schedule. The updated fleet information is then used when creating the schedule for the day.

The proposed method also makes a difference between daily schedule generation and monthly schedule generation. Daily scheduling uses rules and constraints to pick the resources. Monthly scheduling, on the hand, uses a greedy approach that favors buses, with fewer trips assigned and then checks availability. This creates a way to manage schedules without pretending that one single optimization method handles all scheduling choices.

Summary of Scheduling Literature and Identified Research Gaps

Reference	Main Contribution	What Existing Work Addresses	Remaining Gap Relevant to This Research
Mesquita et al. [1]	Review of planning, operation and control of bus transport systems	Vehicle scheduling, driver scheduling and rostering	Scheduling components are presented as interconnected planning problems; limited focus on an event-driven operational workflow
Shen et al. [2]	Survey of driver scheduling approaches	Driver duties, labor constraints and optimization methods	Does not primarily address real-time fleet disruption and automatic schedule regeneration
Transit Network Timetabling Review [3]	Review of modern transit timetabling and scheduling	MILP, heuristics, metaheuristics and dynamic modelling	Real-time scheduling adjustment and integrated operational decision-making remain challenging
Liu et al. [4]	Dynamic bus scheduling	Online adjustment using time-varying information	Focuses on dynamic operational adjustment rather than the complete event-to-next-day scheduling workflow
Gkiotsalitis [5]	Review of bus holding, rescheduling and stop-skipping	Dynamic control and disruption mitigation	Real-time rescheduling introduces complex vehicle and crew schedule interactions

Li et al. [6]	Vehicle rescheduling after breakdowns	Reassignment of vehicles following breakdowns	Strong optimization-based breakdown recovery, but limited integration with a broader monitoring-to-scheduling application workflow
Li et al. [7]	Vehicle rescheduling with retiming	Delaying trips to reduce cancellations after breakdowns	Focuses on optimization of disrupted trips rather than automatic fleet-state-driven subsequent schedule generation
Lai & Leung [8]	Real-time disruption management	Integrated vehicle and crew rescheduling using rolling horizon	Computational optimization focus; practical rule-based resource eligibility and next-day regeneration are less emphasized
Gkiotsalitis & Cats [9]	Review of real-time public transport control	Rescheduling, dispatching and disruption control	Highlights the complexity of real-time operational control and need for practical approaches
Shen et al. [10]	Resilience-oriented timetable optimization	Multi-disruption timetable optimization	Identifies the need for better integration of disruption management and timetable optimization

2. OBJECTIVE

2.1 Main Objective

The main objective of this system component is to develop a time-slot engine that follows predefined transport policies to manage bus operational scheduling through its programmable time slots. The system will classify bus operations into structured time slots such as peak, normal, and night service periods while ensuring buses operate according to approved regulatory policies.

The engine will analyze operational data such as trip start time, route information, and service category to automatically determine the correct operational time slot and apply the relevant transport policies. The system uses automated policy-based scheduling to achieve better service consistency while decreasing manual timetable work and enabling accurate incentive computations for bus operations that meet compliance standards.

- **Specific:** The research investigates transport operational policies through their implementation in different time slots, which include peak, normal, and night service periods.
- **Measurable:** The system achieves 95% accurate results for time-slot classification and policy rule enforcement, which depends on operational trip data.
- **Achievable:** The system operates through a rule-based policy engine, which enables users to set time-slot parameters and process events through automated systems.
- **Relevant:** The solution addresses three problems, which include inconsistent bus scheduling, the absence of standardized operational policies, and challenges in transport regulation enforcement.
- **Time-bound:** The project requires eight months to complete development, integration, and testing for the policy and time-slot engine system.

2.2 Specific Objectives

2.2.1 Policy-Based Time Slot Classification

The first objective is to develop a policy-based time slot classification system, which will classify bus operations according to three predefined service periods: peak hours, normal hours, and night service. The system will use operational data analysis, which includes trip start time and route information and service schedules, to automatically identify the correct operational category.

The classification mechanism will use configurable time-slot definitions and rule-based decision logic to assign each bus operation to the correct service period. The system will implement standardized transport policies to guide scheduling processes and operational monitoring activities. The system needs to reach 95% accuracy for identifying and assigning operational time slots, which will be used for bus services. The classification mechanism will be developed and implemented throughout the initial four months of the research period.

2.2.2 Transport Policy Rule Management

The second objective is to create a transport policy rule management system that enables transport administrators to establish, update, and control their operational procedures for bus services. The system will store policy rules related to service time slots and route operation conditions, and operational constraints within a centralized rule management module.

The policy engine will evaluate bus operations against these predefined rules to ensure compliance with regulatory requirements and approved transport policies. The operational policies will undergo flexible updates through this rule-based framework, which does not need any system modifications. The system needs to enforce policies correctly while it evaluates bus operations through its established rules. The development of the transport policy rule management module will be completed within six months.

2.2.3 Automated Schedule Adjustment Mechanism

The third objective is to create an automatic system that changes bus schedules when operational events and policy rules require schedule adjustments. The system will identify service disruptions, which include delays and missed departures, to automatically assess different scheduling options that fit within established time frames.

The engine uses policy-driven decision logic to find the next available bus or operational window that meets both transport policies and service requirements. The automated adjustment mechanism will sustain service operations while it reduces schedule conflicts that occur during times of operational interruption. The system works to enhance scheduling dependability while it establishes bus operations that will proceed according to existing policies. The development and testing of this mechanism will be completed within seven months.

2.2.4 Policy Compliance Monitoring

The final objective is to develop a policy compliance monitoring module that evaluates bus operations according to established transportation policies and their designated operating times. The system will continuously analyze operational records and scheduling data to detect policy violations or irregular service patterns.

The monitoring system will produce operational reports together with alerts, which provide administrators with information about the status of policy compliance and any potential operational problems. The evaluation process will use performance indicators such as rule compliance accuracy, scheduling consistency, and operational reliability. The system aims to achieve 90–95% accuracy in policy compliance monitoring during testing. The complete implementation and evaluation of this module will be finalized within the eighth month of the project timeline.

3. METHODOLOGY

3.1 Research Approach

The researchers used a Design Science Research approach to create and test the Policy & Time-Slot Engine, which handles transport policy and scheduling operations. The research requires Design Science because its purpose is to create a working system element that provides a solution to actual public transport management challenges.

The study develops a rule-based engine that classifies bus operations into specific time slots while using policy rules to enable automatic scheduling and operational decision processes. The design process requires transport authorities to establish their operational needs while creating flexible policy frameworks and building algorithms that will test bus operations according to those established policies.

The system component will be developed through iterative design and testing phases. The first policy engine prototypes will undergo implementation and improvement based on simulated operational data and testing results. The system performance assessment will measure its ability to accurately identify time slots while applying policy rules and enabling scheduling decisions based on established transport regulations.

3.2 Technical Methods and Algorithm

The system employs a range of technical methods and algorithms distributed across its four functional layers. Each is described below.

3.2.1 Policy Rule Configuration Engine

The system provides a configurable policy rule engine that transport administrators use to establish operating policies, including service times, permitted routes, and scheduling restrictions. The system stores these rules, which the system uses for real-time applications during scheduled operations and policy testing.

3.2.2 Time-Slot Classification Algorithm

The time-slot classification algorithm assigns each bus operation to one of three established service periods, which include Peak service, Normal service, and Night service. The algorithm determines the correct operational category by comparing trip timestamps with the established time intervals.

3.2.3 Policy-Based Schedule Adjustment

The system examines operational windows together with existing policy restrictions when service interruptions result from delayed service or lost service times. The algorithm identifies the next available time slot that meets policy requirements while allowing uninterrupted bus operations.

3.2.4 Policy Compliance Verification

The engine assesses operational records in real time to check their adherence to established transport regulations. The system validates each trip by examining three elements, which include route assignment and scheduled time, and operational regulations, to determine which service operations meet compliance standards.

3.3 Tools and Platforms with Justification

Component	Specification	Justification
ESP32 Microcontroller	Dual-core 240 MHz, built-in Wi-Fi and Bluetooth, 520 KB SRAM	Selected as the central on-board processing unit due to its integrated Wi-Fi capability (eliminating the need for a separate communication module), low power consumption, sufficient processing power for real-time geofence calculations and IMU signal processing, the ESP32 supports the Arduino framework, enabling rapid prototyping.
Arduino IDE	Firmware development for ESP32	Used to write and upload code to the ESP32 device. Provides libraries and tools needed for GPS, sensors, and Wi-Fi communication.
Azure IoT Hub	IoT device communication and data ingestion	Allow ESP32 devices to securely send real-time data to the cloud. It manages communication between many IoT devices and the backend system.
Spring Boot	Backend REST API and business logic	Used to build the backend server that processes device data, manages APIs, and connects the database with the frontend applications.
PostgreSQL	Primary database	Stores system data such as bus routes, locations, drivers, and logs in a structured and reliable way.
React	Regulator web dashboard	Used to create an interactive web dashboard for monitoring bus locations, routes, and system data.

Table 1- Components & Justifications

Tool	Purpose	Justification
Git / GitHub	Version control and collaborative development	Used to track code changes and allow multiple developers to work on the project efficiently.
Postman	API endpoint testing and documentation	Helps test backend APIs and verify requests and responses during development.
Arduino Serial Monitor / Arduino Serial Plotter	Real-time IMU and GPS data debugging during firmware development	Used to view and debug sensor data such as GPS coordinates and motion readings from the ESP32 device.
Google Maps Roads API	Route polyline verification and road-snapping	Used to validate and align bus routes with real road paths for accurate route mapping.

Table 2- Tools & Justifications

3.4 Data Collection and Ethical Considerations

3.4.1 Data Collection Methods

The Policy & Time-Slot Engine evaluation process requires data from multiple sources to verify the correct functioning of policy rules and time-slot classification and scheduling decisions. The system generates operational trip records, which serve as the primary data source for the assessment of trip start time and route identification, service duration, and scheduling information. The operational records enable researchers to examine how the policy engine categorizes bus operations into three predefined service periods, which include peak hours and normal hours, and night service.

The research team creates simulated datasets that serve as an additional source for evaluating policy engine performance across various transport scenarios. The simulations simulate peak-hour congestion conditions together with delayed departures, missed trips, and overlapping service schedules. The system uses simulated data to evaluate rules and time-slot classifications and automate scheduling changes without depending on actual operational data.

The Western Province Road Passenger Transport Authority provides official transport policy documents together with route information, which serve as reference data to establish operational policies and service time-slot definitions. The policies establish approved operating hours and service frequency regulations, and route management guidelines. Testing requires additional geographic route data from OpenStreetMap, which helps identify routes and provide operational context.

3.4.2 Ethical Considerations

The system requires vehicle tracking and driver monitoring, which requires multiple ethical standards to be implemented. Testing participants receive project information, which allows them to choose whether to participate or not. The system uses anonymous labels to identify vehicles and drivers because it does not keep any personal data. The IoT device establishes encrypted links with the backend server for secure data transmission, while only authorized users can access the database. The system operates with its essential operational functions by collecting only necessary data, which it processes through device-based raw sensor data handling. The project follows Sri Lanka Institute of Information Technology research ethics guidelines, which require formal approval for upcoming tests that involve external bus operators. The organization uses driver behavior data to create safer driving conditions through positive feedback mechanisms, which they use for monitoring purposes.

3.5 Validation Strategy and Performance Metrics

The Policy & Time-Slot Engine validation process uses structured testing methods to measure three elements, which include policy rule evaluation accuracy, time-slot classification accuracy, and automated scheduling decision accuracy. The system will undergo testing through two different methods, which include testing with simulated operational datasets and testing with real operational records obtained during system testing. The datasets contain various transport scenarios, which include peak-hour operations and normal service periods, night services, and unexpected scheduling disruptions.

The primary validation process focuses on evaluating the accuracy of the time-slot classification algorithm. The system will analyze operational timestamps and compare the assigned service categories with predefined policy definitions to determine classification accuracy. Another major assessment point focuses on policy rule enforcement, which requires the system to determine whether bus operations follow the established transport policies and service schedules.

The system uses various performance metrics to evaluate its operational efficiency. The system assesses time-slot classification accuracy through its ability to correctly divide operations into peak hours and normal service times, and night service periods. The policy rule enforcement accuracy measures how well the engine detects bus operations that meet or break the established policy rules.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed solution uses a multi-layered IoT-to-cloud system, which enables bus operation monitoring and route compliance enforcement, policy-based scheduling, and incentive management. The onboard IoT device, which exists on each bus, functions to acquire GPS data and timestamp information and detect motion events. The communication layer transmits event packets to the cloud platform, which executes its processing functions. The cloud processing layer uses multiple core modules that work together to perform route monitoring, map matching, and trip segmentation and distance calculation, revenue computation, and the Policy & Time-Slot Engine. The component function detects trips according to three defined service periods, which include peak, normal, and night, while it uses transport policies to make scheduling and compliance choices. The presentation layer provides outputs that users can access through a regulatory web dashboard and a bus owner mobile application to enable them to track operations, assess policy results, and make decisions.

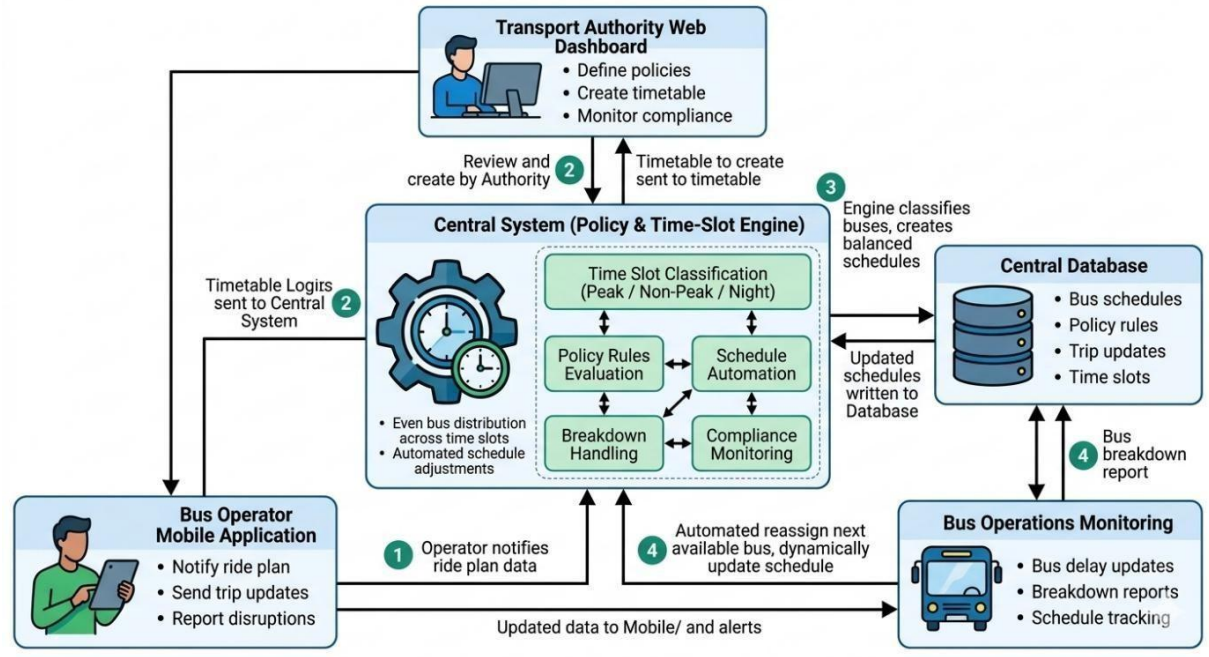


Figure 1- High-Level System Architecture

5. USER REQUIREMENTS

5.1 Stakeholder identification

Multiple stakeholders use the Policy and Time-Slot Engine system to ensure that bus operations follow approved transport policies and schedules. The primary stakeholders at transport authorities hold responsibility for creating operational policies, managing time-slot rules, and ensuring regulatory compliance. The system serves bus operators and owners who need to access service schedules and policy information that impacts their operational activities and incentive programs. Timekeepers use the system to monitor departures and verify whether buses operate within the assigned time slots. Bus drivers indirectly participate in the system because they operate according to the schedule that the policy engine produces. The stakeholders work together to establish regular bus schedules and enforce policies that lead to trustworthy public transportation services.

5.2 Functional Requirements

FR1	User Registration and Login
Input	User credentials (Route Permit No / Username, Password)
Processing	Validate credentials, encrypt password, authenticate user, and manage user session.
Output	Successful login confirmation or error message.

Table 3 - FR1

FR2	Real-Time Bus Location Tracking
Input	GPS coordinates from the bus tracking device
Processing	Receive GPS data, process location information, update system database
Output	Real-time bus location displayed on the monitoring dashboard

Table 4- FR2

FR3	Time-Slot Classification
Input	Bus trips start time, route ID, and predefined time-slot definitions (peak, normal, night).
Processing	The Policy & Time-Slot Engine compares the trip start time with configured time-slot ranges and evaluates applicable policy rules to determine the correct operational category.
Output	Assigned time-slot category for the bus operation and confirmation that the trip belongs to the defined service period.

Table 5- FR3

FR4	Trip Start and End Monitoring
Input	Trip start/end confirmation from the mobile app or GPS triggers
Processing	The system detects trip initiation and completion events and updates trip status in the database
Output	Trip status updated and displayed in the operational dashboard

Table c- FR4

FR5	Automated Schedule Adjustment
Input	Detected service disruption events such as delayed departure, missed trip, or unavailable bus.
Processing	The system analyzes the disruption event and evaluates available time slots and policy constraints to identify the next valid operational schedule.
Output	Recommended alternative time slot or updated schedule that satisfies operational policies and maintains service continuity.

Table 7- FR5

FR6	Notification and Alert System
Input	Route deviation alerts, system status data, and high-speed
Processing	Analyze system events and trigger notifications
Output	Alerts for operators regarding route violations or high-speed

Table 8- FRc

FR7	Admin Web Dashboard
Input	Admin login credentials
Processing	Authenticate admin and allow CRUD operations
Output	Web-based dashboard with data management features

Table S- FR7

5.3 Non-functional Requirements

NFR1	**System Reliability**
Description	The system should operate continuously with minimal downtime to ensure uninterrupted monitoring of buses.

Table 10- NFR1

NFR2	**Data Security**
Description	The system must protect sensitive information such as user credentials and financial data using secure authentication and encryption mechanisms.

Table 11- NFR2

NFR3	**Performance**
Description	The system should process GPS tracking and ticketing data quickly to provide real-time updates without delays.

Table 12- NFR3

NFR4	Usability
Description	The system interfaces should be easy to use for bus owners and transport authorities, with minimal training required.

Table 13- NFR4

NFR5	Usability
Description	The system interfaces should be easy to use for bus owners and transport authorities, with minimal training required.

Table 14- NFR5

NFR6	Scalability
Description	The system should support an increasing number of buses, users, and transactions as the transportation network grows.

Table 15- NFRc

6. COMMERCIALIZATION PLAN

The proposed IoT-based smart bus route monitoring system has strong potential to be developed as a practical digital solution for the public transportation sector. The system enables bus services to achieve greater operational dependability through its capabilities of monitoring routes in real time, enforcing policies automatically, and scheduling through time slots. The Policy & Time-Slot Engine component enables transport authorities to enhance their bus operation management through its ability to automate scheduling tasks that were done manually by enforcing predefined policies. Public transport management will benefit from improved operational transparency, which will decrease service interruptions and enhance decision-making capabilities.

Target Market and Customer Persona

The primary target market for this system includes government transport authorities, regional passenger transport boards, and regulatory bodies that oversee bus operations. The system will assist Sri Lankan institutions, such as the Road Passenger Transport Authority, to monitor route compliance and enforce operational policies. Private bus operators and bus owners represent secondary customers who will use the system to monitor their operational performance while ensuring they meet transport regulations.

Value Proposition

The primary value of the system brings together IoT-based monitoring systems with automatic operational management capabilities. The Policy & Time-Slot Engine enables automatic classification of service periods while it controls transport policies and makes schedule changes during service interruptions. This technology decreases the need for human monitoring operations while enhancing the reliability of bus service delivery.

Cost Estimation and Pricing Strategy

The system requires major expenses for three areas, which include IoT device installation, cloud infrastructure, software development, and system maintenance. The system operational costs will stay low because the core system functions as a software-based platform after its initial launch. The system provides two options for transportation authorities, which include a government-supported transport management platform and a licensed transport agency solution.

Revenue Model

The revenue streams for the business will derive from three different sources, which include government licensing agreements and subscription services for transport authorities and service contracts that provide bus fleet monitoring.

Competitive Advantage and Intellectual Property Considerations

The system achieves its main competitive edge through its unique capability to integrate policy-based scheduling systems with automated operational monitoring functions, which current transport monitoring systems do not offer. The research outcomes include two components, which emerge from the proposed scheduling algorithms and policy-driven operational management mechanisms that researchers can use to establish intellectual property rights.

7. BUDGET AND JUSTIFICATION

The estimated project cost includes software development, hardware infrastructure, networking services, and maintenance expenses required to deploy the proposed Smart Bus Route Monitoring and Revenue Management System. The budget reflects the costs associated with building a scalable real-time transport monitoring platform capable of supporting multiple buses and operational users.

DESCRIPTION	AMOUNT (LKR)	TOTAL (LKR)
SOFTWARE		
React.js	6 000	
Spring Boot	10 000	
React native	10 000	
C++	7 000	
PostgreSQL	9 000	
Visual Code	5 000	
Software Cost		47 000
HARDWARE		
Router	6 000	
2 pc	40 000	
IoT Device	9 000	
Total Hardware Cost		55 000
NETWORK		
Data transfer and API usage	8 000	
Hosting and server deployment	7 000	
Total Networking Cost		15 000
MAINTENCE		
Client training and technical support	6 000	
Warranty and support services	10 000	
Security software and monitoring	4 000	
System maintenance agreement	5 000	
Total Maintenance Cost		25 000
Total Project Cost		**142 000**

Table 1c- Budget Calculation

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) provides a hierarchical representation of all tasks required for the predictive warfarin dosing system. The project breaks down into its main phases, which include requirement analysis, system design, data collection model development, implementation testing, deployment, and documentation. Each phase is further broken down into specific, manageable tasks, ensuring clarity in responsibilities and deliverables. WBS enables project teams to monitor their work as they distribute resources, manage project timelines, and identify potential risks. The project structure enables stakeholders to track project progress while upholding quality standards and verifying that the system achieves both clinical and operational goals.

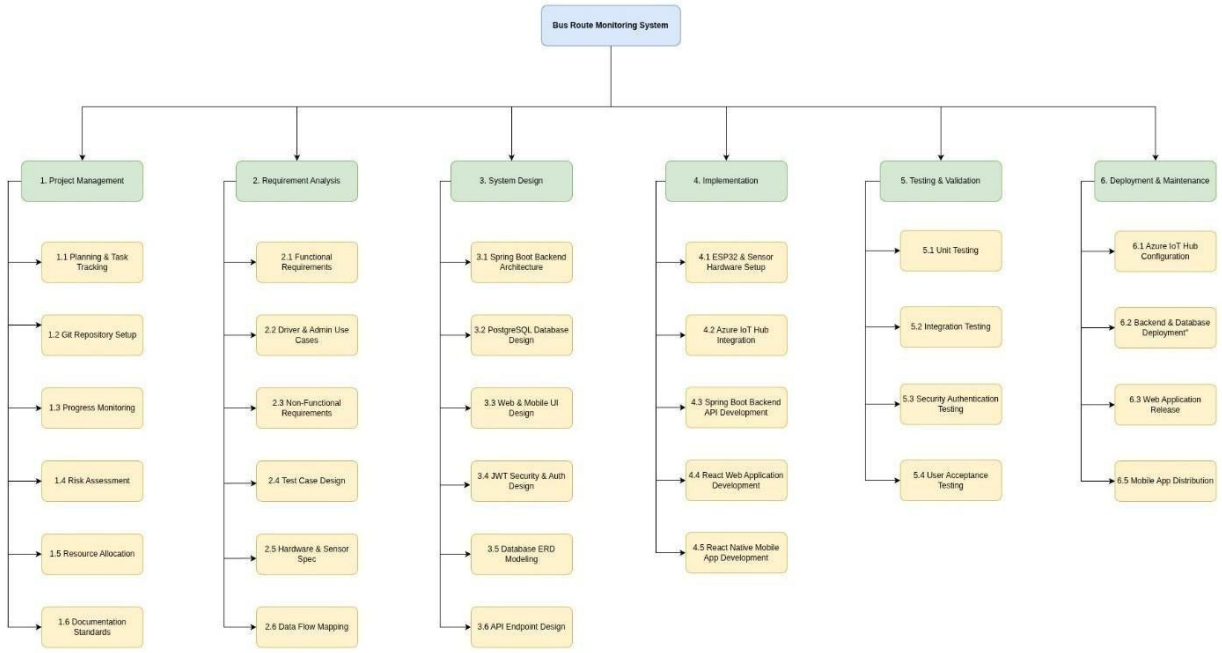


Figure 2 - Work Breakdown Chart

The Work Breakdown Structure (WBS) outlines the primary tasks necessary to complete the research project.

Task ID	Task Description	Responsible Member	Timeline
T1	Literature review and problem analysis of public bus transport monitoring systems	All Members	Weeks 1-3
T2	Collection of transport operational data, route schedules, and policy information from the Road Passenger Transport Authority	All Member	Weeks 3-5
T3	Development of an IoT-based GPS event monitoring and trip data acquisition module	Member 1	Weeks 5–7
T4	Development of route compliance verification and map-matching algorithm	Member 1	Weeks 7–10
T5	Design and implementation of Policy & Time-Slot Engine for automated scheduling and service classification	Member 2	Weeks 8–11
T6	Development of an incentive and revenue calculation module based on operational distance and service periods	Member 3	Weeks 9–12
T7	Database design and implementation for route data, policy rules, and operational logs using PostgreSQL	All Members	Weeks 10–12
T8	Development of a regulator web dashboard using React for monitoring routes and policy compliance	Members 2 & 3	Weeks 11–13
T9	Development of mobile applications for bus operators to update trips and receive operational notifications	Member 4	Weeks 11–13
T10	Integration of IoT monitoring, policy engine, revenue module, and user interfaces	All Members	Weeks 14–16
T11	System testing and validation using simulated and real operational scenarios	All Members	Weeks 16–18
T12	Performance evaluation, documentation, and final system improvements	All Members	Weeks 18–20

9. GANTT CHART

The Gantt chart shows both the research project activities schedule and their planned execution order. The document defines main project stages, which include literature review and dataset collection, model development, system implementation and testing, and final documentation. The project schedule divides tasks into multiple months to achieve continuous development and complete project milestones on time. The structured timeline enables progress monitoring of development work while ensuring both separate components and the complete system are finished during the designated research timeframe.

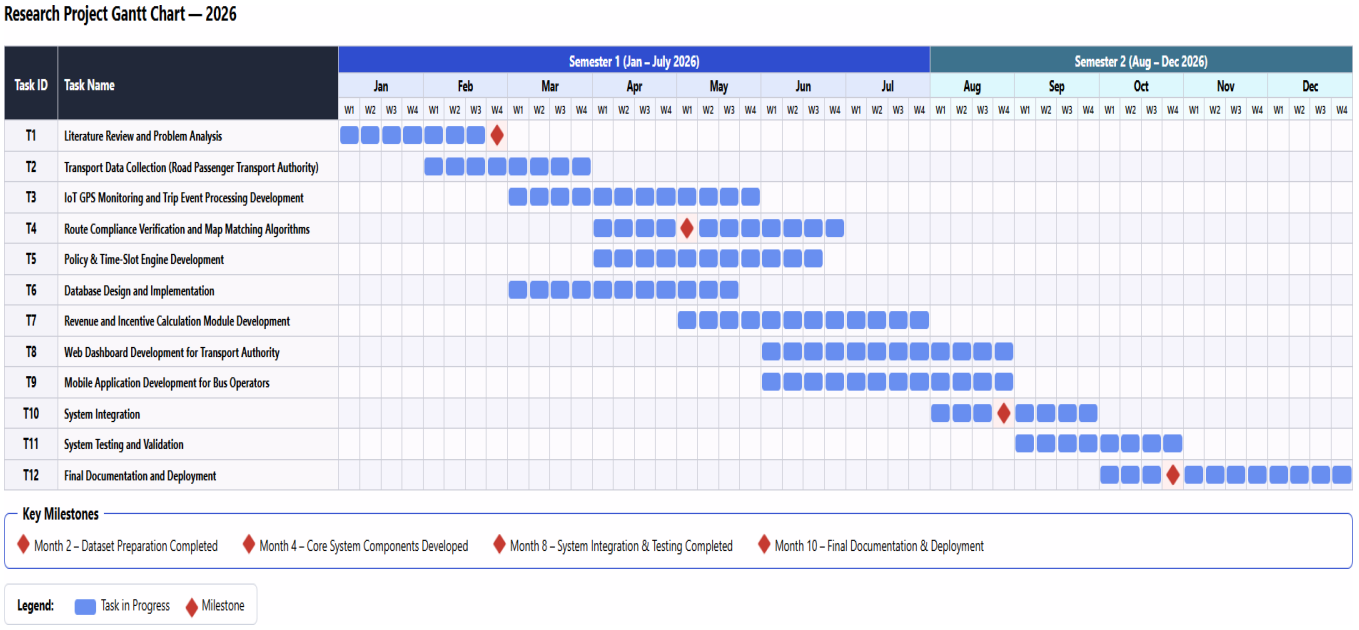


Figure 3- Gannt Chart

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APPENDICES

Appendix A – Survey / Interview Instrument

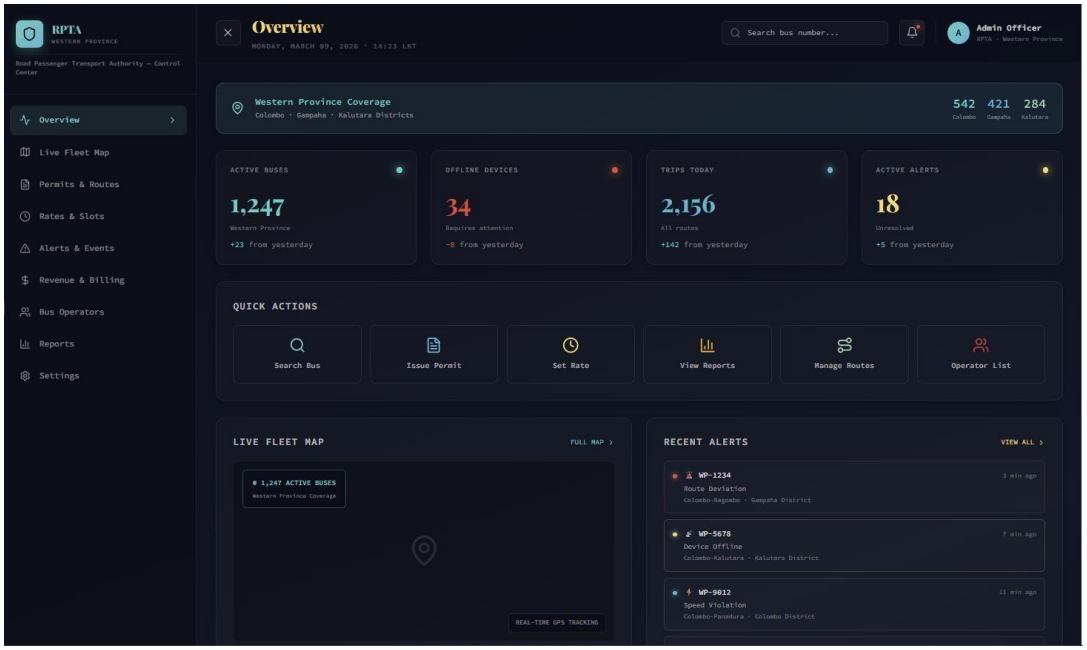
The appendix contains information collected through stakeholder discussions and consultations on public transport operations. The research data and insights were primarily derived from the Road Passenger Transport Authority, which included route tracking data, scheduling information, and details about bus operation difficulties shared by the system administrator.

Appendix B - Detailed System Diagrams

The appendix presents complete technical diagrams that were utilized throughout the system development process. The system consists of three main components, which include overall system architecture, component diagrams, and process flow diagrams to demonstrate how the IoT monitoring devices work with the cloud processing layer and Policy & Time-Slot Engine in the system.

Appendix C - Preliminary UI mockups

Sample UI for Web Application Dashboard



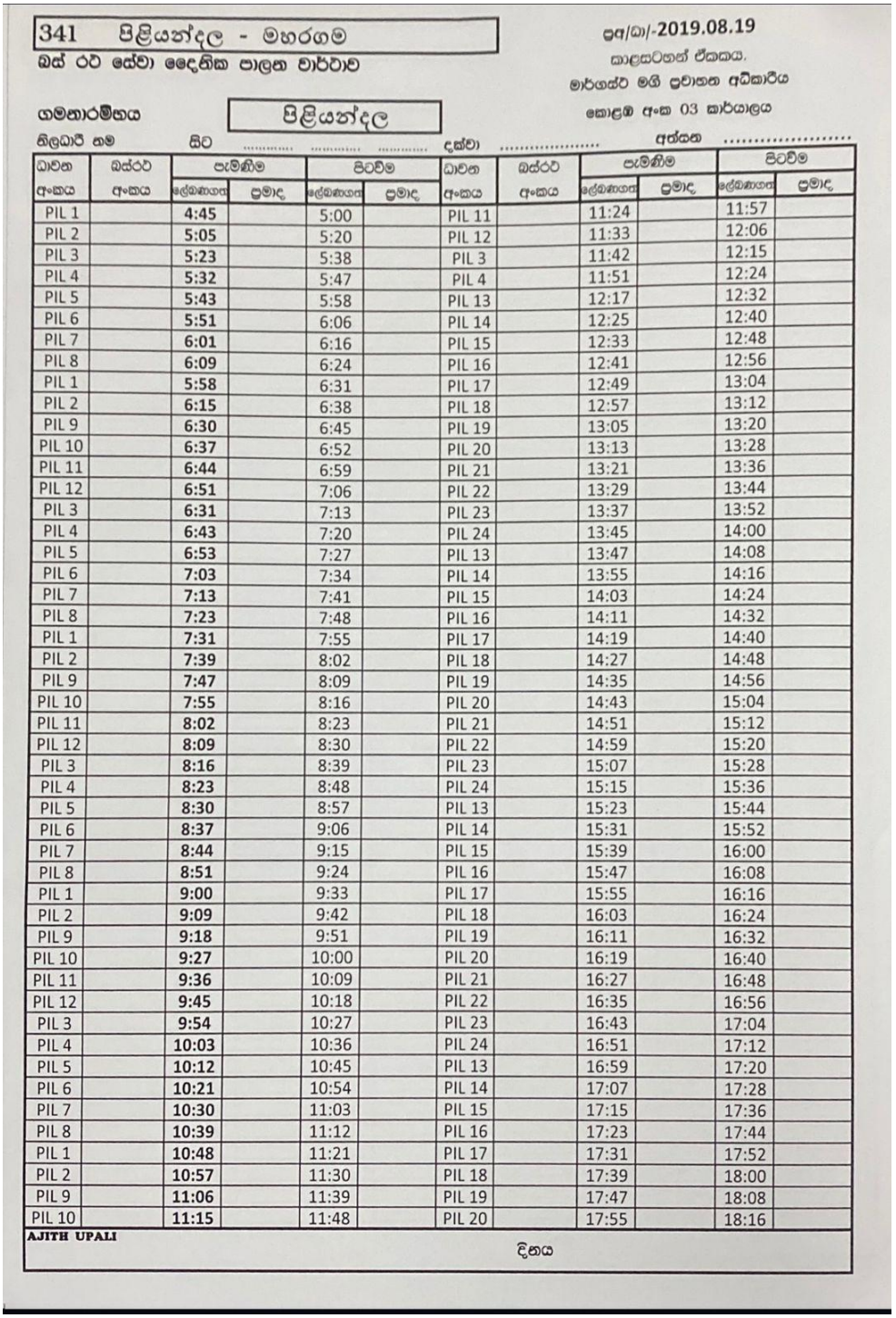
Appendix D - Risk Analysis

The appendix presents an overview of risks that may arise during the execution of the suggested system. The identified risks for this project include two specific hazards, which are GPS signal inaccuracies and communication interruptions that affect IoT data transmission. The system needs to face the difficulties

of connecting its different elements and of collecting authentic operational data from actual environments. The document presents possible methods to reduce the identified dangers.

Appendix E - Ethical documents

Current use bus schedule



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